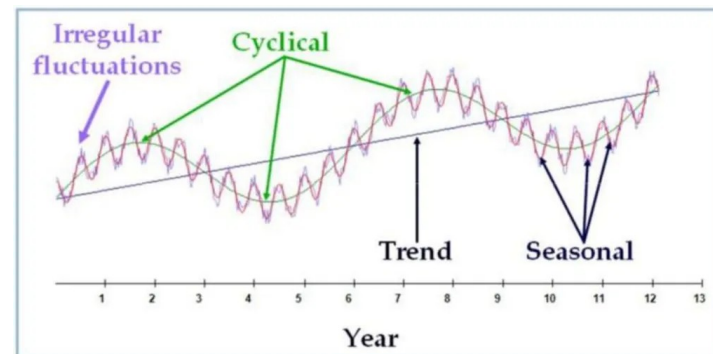


Auto-Regression

Management Science Principles

Recap

- Last lecture we covered time series data:
 - Attributes of time series
 - Basic forecasting techniques
 - Naïve baseline
 - Moving averages
 - Practiced with Excel
- Let's look at some additional forecasting methods, this time applied in Python!



Moving Average Models

Moving Average Model

- In signal processing, a moving average model is used to predict future values based on the **error** or “white noise” output of the model.

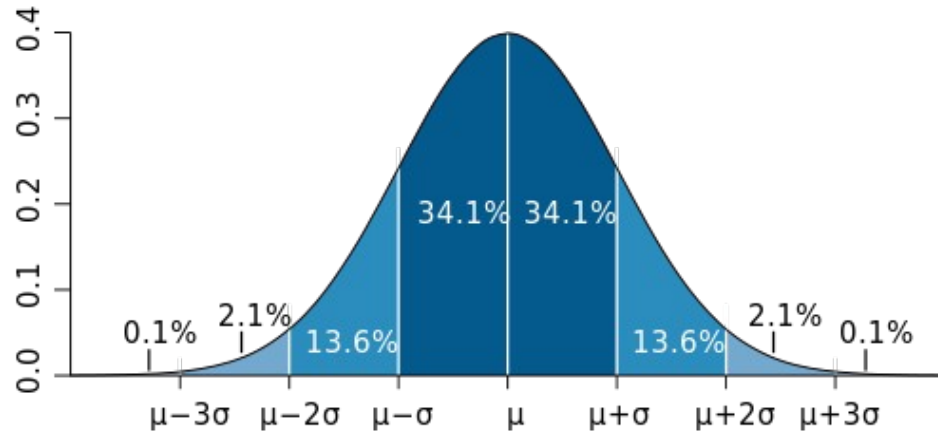
$$f_t = \mu + \phi_1 \epsilon_{t-1}$$

Note: In math- or signal-focused domains, this may be referred to as moving average process, just as other models we'll see may be called a process.

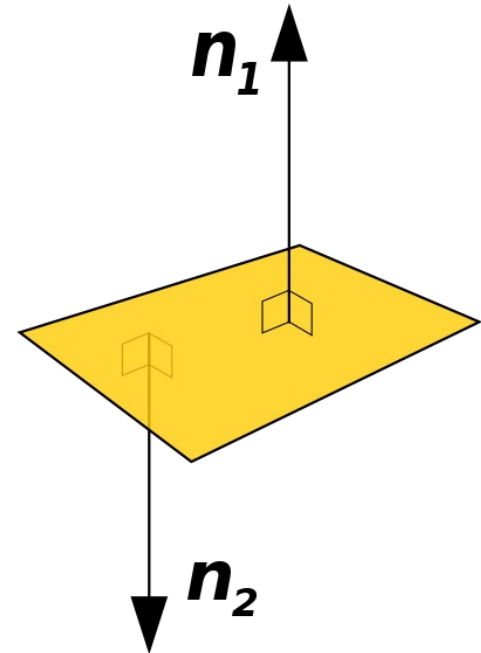
Moving average models vs. Moving averages?

- Indeed, we just talked about moving averages in the context of averaging the actual values over defined windows in time!
- Why is this technique based on using errors of values over time named the same?
 - Especially confusing since it's just a linear combination of error terms like a weighted moving average would be of actual values.

Normal

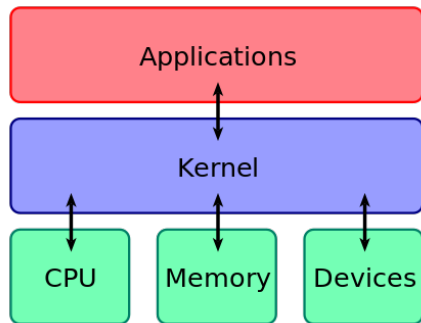


Normal

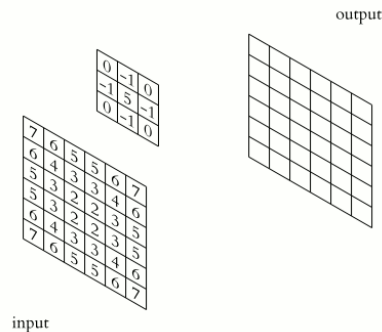


<https://en.wikipedia.org/wiki/Normal>

Kernel



Kernel



Kernel

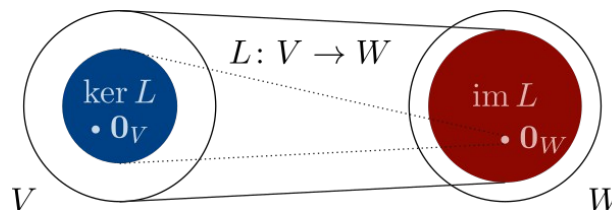
```
// Kernel function to add two arrays __kernel
void vectorAdd(__global const float* a,
__global const float* b, __global float*
result, const int size) { // Get the index of the
current work-item int index =
get_global_id(0); // Check if the index is within
the bounds of the array if (index < size) { //
Perform the addition result[index] = a[index] +
b[index]; }
```

Kernel

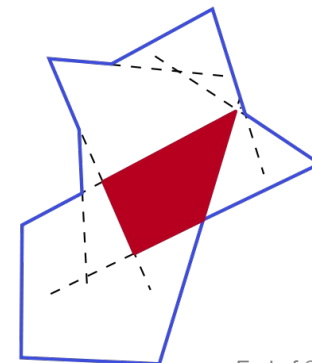
The **Fisher kernel** is defined as

$$K(X_i, X_j) = U_{X_i}^T \mathcal{I}^{-1} U_{X_j}$$

Kernel



Kernel



<https://en.wikipedia.org/wiki/Kernel>

Unfortunately, names are not always meaningful; they may just be popular.

- From a [Stack Overflow discussion](#) on moving average nomenclature, an excerpt from Alan Pankratz's book *Forecasting with Univariate Box-Jenkins Models* states this:
 - “The label “moving average” is technically incorrect since the MA coefficients may be negative and may not sum to unity. This label is used by convention.”
- Despite the confusing name, the signal processing community has kept “moving average model” or “moving average process” even as many spaces that used “moving averages” have adopted more of these mathematical and signal-based techniques.

Moving Average *Model* Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- Given a process with $\mu = 10$ (average value of 10), we will use the following formula to make forecasts based on the reported error:

$$f_t = \mu + \phi_1 \epsilon_{t-1}$$

t	f_t	e_t	y_t
1			
2			
3			
4			
5			

Moving Average Model Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- First, with no error term, we simply use the mean to forecast the initial value:

$$f_1 = 10$$

t	f_t	e_t	y_t
1	10		
2			
3			
4			
5			

Moving Average Model Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- Once we have an error term, we can make forecasts including that term.
- Part of the task of building this model is to identify the weights, i.e., ϕ_1 in this case, that optimally fit our data.
 - For the purposes of showing how the model works, in this example we will just select and use $\phi_1 = 0.5$

$$f_2 = \mu + \phi_1 \epsilon_1$$

$$\textcolor{red}{!} 10 + 0.5 * -2 = 9$$

t	f_t	e_t	y_t
1	10	-2	8
2			
3			
4			
5			

Moving Average Model Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- The model can progress through time using the same calculations:

$$f_3 = \mu + \phi_1 \epsilon_2$$

$$\textcolor{red}{i} 10 + 0.5 * 1 = 10.5$$

t	f_t	e_t	y_t
1	10	-2	8
2	9	1	10
3			
4			
5			

Moving Average Model Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- The model can progress through time using the same calculations:

$$f_4 = \mu + \phi_1 \epsilon_3$$

$$10 + 0.5 * 0 = 10$$

t	f_t	e_t	y_t
1	10	-2	8
2	9	1	10
3	10.5	0	10.5
4			
5			

Moving Average Model Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- The model can progress through time using the same calculations:

$$f_5 = \mu + \phi_1 \epsilon_4$$

$$10 + 0.5 * 2 = 11$$

t	f_t	e_t	y_t
1	10	-2	8
2	9	1	10
3	10.5	0	10.5
4	10	2	12
5			

Moving Average Model Example (credit: <https://www.youtube.com/watch?v=voryLhxiPzE>)

- We can imagine the model continuing through time...
- That was just one example of an MA(1) model using a single time point of the past.
- Generically, we refer to an MA(q) model as a q-order model defined as below, where the parameters are weights ϕ_i :

$$f_t = \mu + \epsilon_t + \sum_{i=1}^q \phi_i \epsilon_{t-i}$$

t	f _t	e _t	y _t
1	10	-2	8
2	9	1	10
3	10.5	0	10.5
4	10	2	12
5	11	1	12

ϵ_t may be called “white noise” in the signal processing space, but **bias** maybe a better descriptor to give ML engineers since it is used similarly.

Why is this called a moving average model?

- It is applied over a time series which is assumed to have a defined mean (μ).
- The data is centered around or averages around the mean over time.



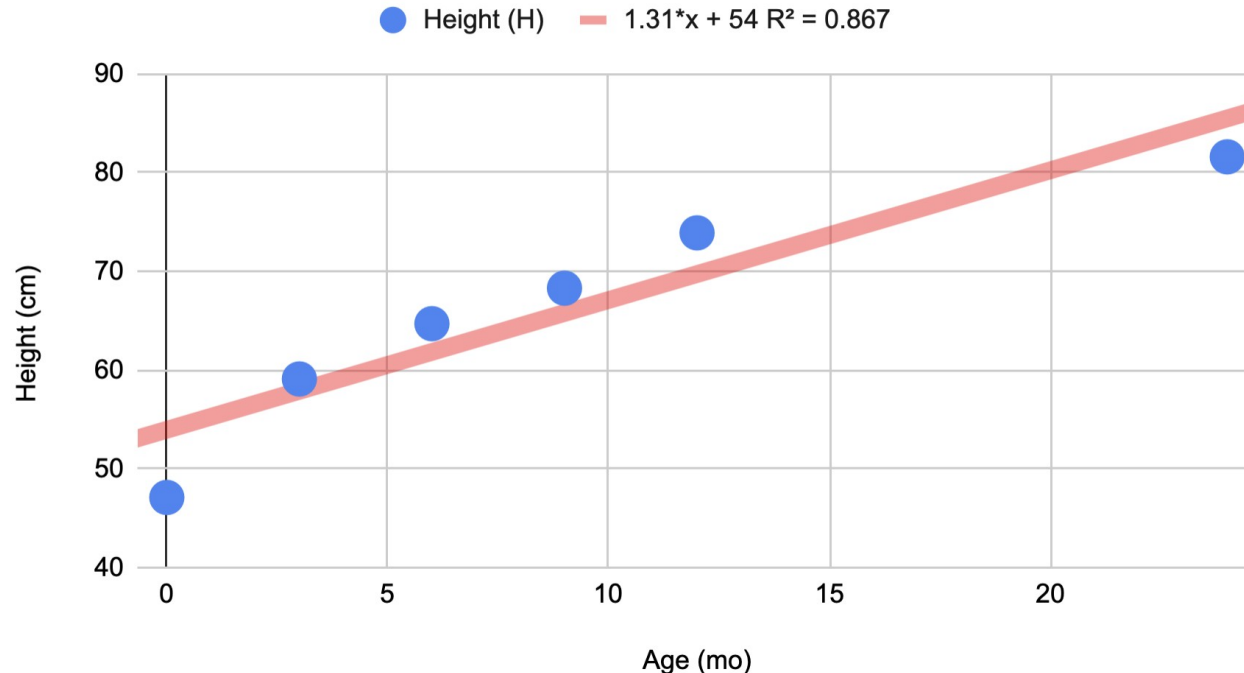
Why use the error term?

- The error itself is another random variable!
 - We could say that it is normally distributed with a mean of 0 and some standard deviation.
- The moving average model gives us a method to forecast based on this random variable.
- What if we **do** want to use the actual values?
 - Perhaps we should just use a moving average like before?
 - Let's introduce another time series data model: auto-regression!

Auto-Regressive Models

Recall Regression Models

- In 270, we introduced our first basic data model as linear regression.
- Linear regression uses one variable to predict another.



What about time series data?

- We only have one variable itself over time.
- Auto-Regression!
 - Use a time-lagged version of the variable to predict *itself*. The “auto” part of the regression.

Market Summary > NVIDIA Corp

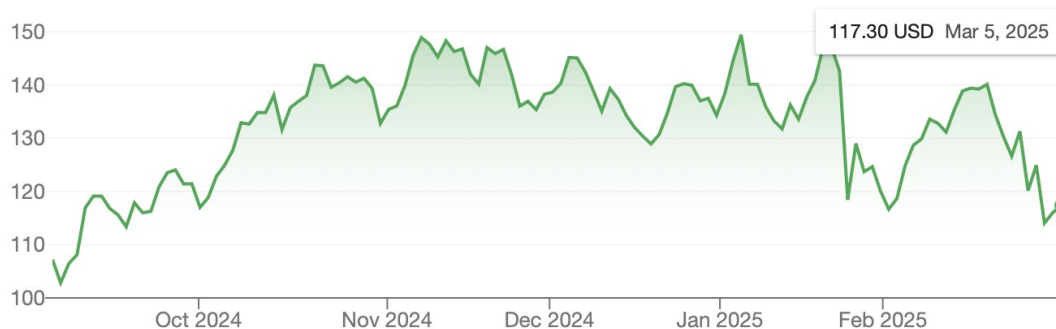
117.30 USD

+10.09 (9.41%) ↑ past 6 months

Closed: Mar 5, 7:47 PM EST • Disclaimer

After hours 116.75 -0.55 (0.47%)

1D | 5D | 1M | **6M** | YTD | 1Y | 5Y | Max



Auto-Regressive Models

- Let's define the p-order auto-regressive model using historical data y :
 - Note that we also include a white noise or bias term ϵ_t

$$f_t = \epsilon_t + \sum_{i=1}^p \theta_i y_{t-i}$$

Above, I'm maintaining the previous terminology of f being our forecast value and y being our actual value.

However, if we simply define it in terms of a time series of data X as shown right, you see the “auto-regressive” part.

It is like a regression model but instead of predicting one variable from another, we predict this variable from itself.

$$X_t = \epsilon_t + \sum_{i=1}^p \theta_i X_{t-i}$$

Auto-Regressive Models

- We've already seen this!
 - The weighted moving average we did in a previous lecture could be considered a very simple autoregressive model with the bias and all weights set to 0 beyond the last 3 samples.
- The main difference is that autoregressive models have no requirement on weight values, while a standard weighted moving average constrains the sum of weights to 1 to ensure the units stay consistent with the original data.

$$f_t = \epsilon_t + \sum_{i=1}^p \theta_i y_{t-i}$$

2000	1451		
2001	1499		
2002	1686		
2003	1764	$= 1/3 * \text{B2} + 1/3 * \text{B3} + 1/3 * \text{B4}$	

Auto-Regressive Moving Average Model (ARMA)

Auto-Regressive Moving Average Models (ARMAs)

- We can combine both models to create a new one incorporating actual values and error:

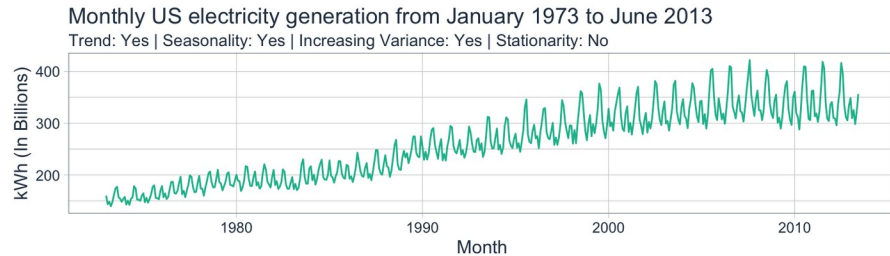
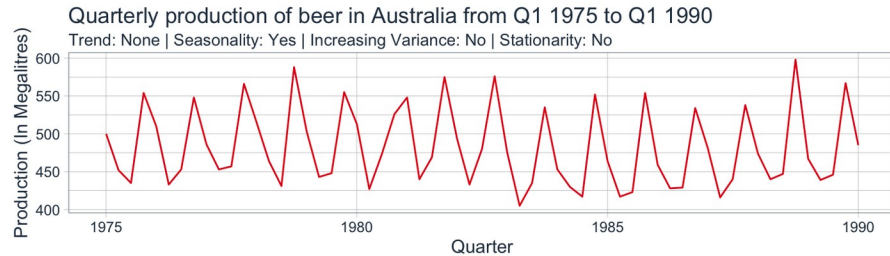
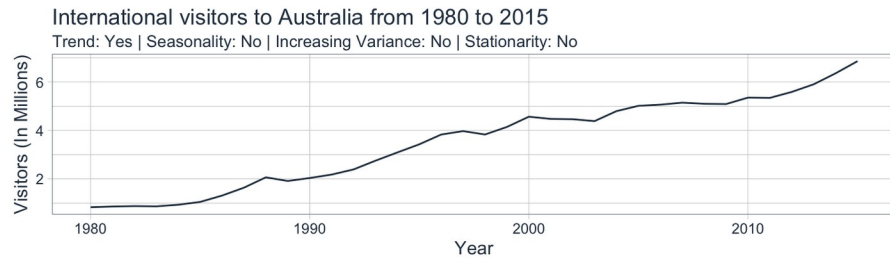
$$f_t = \epsilon_t + \sum_{i=1}^p \theta_i y_{t-i} + \sum_{i=1}^q \phi_i \epsilon_{t-i}$$

- This is called the [Auto-Regressive Moving Average \(ARMA\)](#) model.
 - Funnily enough, it is a seemingly natural extension of the moving average approaches we discussed last time, but the discipline of origin is different with the terminology being completely flipped.

ARMA Example in Python

ARMA and Stationary Data

- Just as we saw with the moving average model assuming the data was centered around the same mean over time, the ARMA model also assumes stationary data.
 - Stationary data assumes the mean and variance do not vary across time.
- Clearly, this doesn't work well for all types of time series data!
- As we've seen, time series data, like stock prices, fluctuate with seasonality and have trends over time.



Auto-Regressive **Integrated** Moving Average (ARIMA) model

- The ARIMA model integrates a differencing mechanic to make non-stationary data stationary.
- In essence, this model zero-centers the data as it goes by subtracting out previous time-lagged values.
- This step enables it to track better with non-stationary data like stocks.



ARIMA Parameters

- As before, parameters p and q are for the number of weighted values to use in the forecast and the error terms, respectively.
- ARIMA adds a parameter d to specify the number of times to difference the data.
 - Tests like the [Augmented Dickey-Fuller test](#) can be used to assess whether a time series is stationary or not, which can be used to determine degree of differencing, d , for ARIMA.
 - See the [Time Series Analysis Handbook](#) for more info.

Parameter	Component	Definition
p	Auto-Regression	The number of lags to include in the model AKA the lag order.
d	Integrated	The number of times the data has been differenced to make it stationary AKA the degree of differencing.
q	Moving Average	The number of past errors to include in the model, i.e., the size of the moving average window AKA the order of moving average.

ARIMA Example in Python

Seasonal Auto-Regressive Integrated Moving Average (SARIMA) model

- A lot of time series data includes seasonality. SARIMA models are identical to ARIMA with an entire set of additional terms for seasonal values.
- We add parameters P, D, and Q that are the same as in ARIMA but only correspond to seasonal lags, differencing, and error.
- An important parameter to know in SARIMA is the s, or seasonal period, parameter.

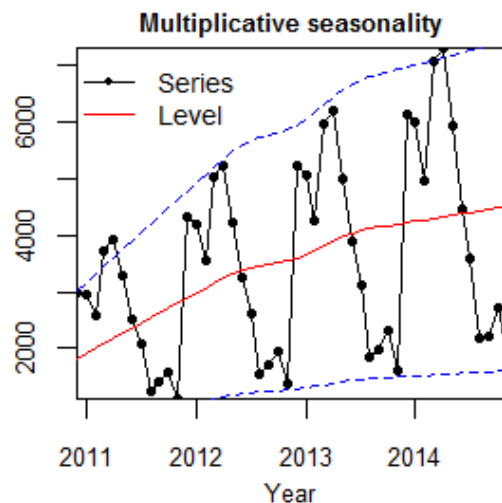
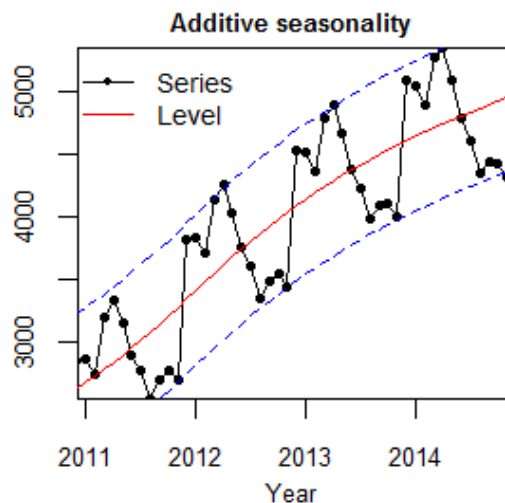
Parameter	Component	Definition
P	Seasonal Auto-Regression	The number of seasonal lags to include in the model AKA the seasonal lag order.
D	Integrated	The number of times the data has been seasonally-differenced to make it stationary AKA the seasonal differenced order.
Q	Seasonal Moving Average	The number of past seasonal errors to include in the model i.e. the size of the moving average window AKA the seasonal moving average order.
s	Seasonal Period	The number of time steps for a single seasonal period, e.g., 7 for weekly seasons of daily data.

SARIMAX in Python

- SARIMAX extends SARIMA further by adding eXogenous variables, which are just any external factors.
 - Notably, this can make the model multi-variate as you can add lists of other values alongside your time series data.
- We'll be using the **statsmodels** library in Python, which includes a flexible SARIMAX implementation which can be anything from ordinary ARMA to its extended forms.
 - www.statsmodels.org/dev/generated/statsmodels.tsa.statespace.sarimax.SARIMAX.html
 - www.statsmodels.org/dev/examples/notebooks/generated/autoregressions.html

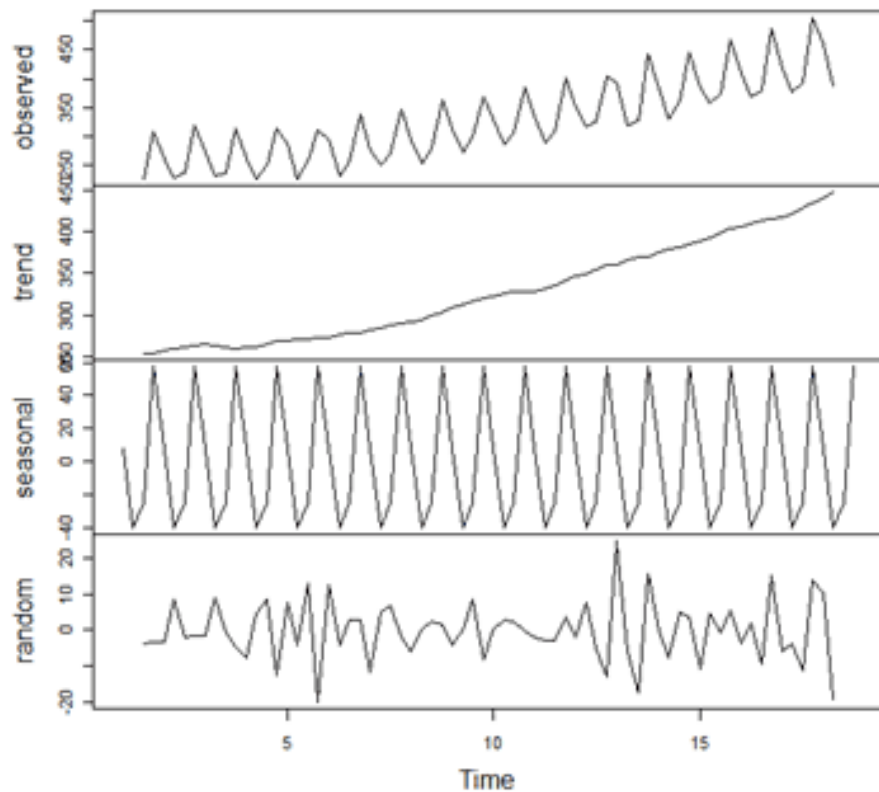
Decomposing Time Series Data

- There is an entire field focused on time series decomposition!
- Two main structures used in basic decomposition models:
 - Additive: Time Series = Trend + Seasonal + Irregular
 - Multiplicative: Time Series = Trend * Seasonal * Irregular



Credit: <https://kourentzes.com/forecasting/2014/11/09/additive-and-multiplicative-seasonality/>

Decomposition of additive time series



Credit: <https://online.stat.psu.edu/stat510/book/export/html/667>

Additional Resources

- We are covering multiple topics and models at a high-level primarily for implementation purposes (although occasionally diving into the math), but I encourage you to explore deeper into the math and statistics as desired!
- Good resources include:
 - [Time Series Analysis Handbook](#)
 - [Penn State's STAT 510 course](#)
- Further useful references:
 - <https://towardsdatascience.com/forecasting-time-series-data-stock-price-analysis-324bcc520af5>
 - <https://www.joonim.net/2021/02/01/tidy-forecasting-in-r-arma-models/>